

Robust Offline Safe RL with Soft OOD Cost Regularization

A Small-Scale Extension of CCAC

RL Final Project Presentation

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Background: Offline Safe RL

Offline safe reinforcement learning learns a policy from a fixed dataset while controlling cumulative safety cost.

$$\max_{\pi} \mathbb{E}_{\pi}[R(\tau)] \quad \text{s.t.} \quad \mathbb{E}_{\pi}[C(\tau)] \leq \epsilon$$

The setting is difficult because the policy must handle:

1. distribution shift from offline data
2. safety constraints during evaluation
3. different or changing safety budgets

CCAC as the Backbone

CCAC learns a constraint-conditioned actor-critic policy.

$$\pi_{\theta}(a_t \mid s_t, \kappa_t), \quad \kappa_{t+1} = \kappa_t - c_t$$

The remaining budget κ_t is part of the policy input, so one trained policy can be evaluated under multiple target costs.

Component	Role
CVAE	generates budget-conditioned state-action samples
OOD classifier	identifies generated OOD / unsafe samples
cost critic	overestimates cost for risky generated samples
actor	maximizes reward while avoiding high predicted cost

Motivation: Brittle OOD Decisions

The original CCAC implementation uses a hard OOD decision:

$$\mathbb{I}_{\text{OOD}}(s, a, \kappa) = \mathbb{I}[h_{\psi}(s, a \mid \kappa) > 0.5]$$

This creates two safety-relevant risks:

1. near-threshold samples such as 0.49 and 0.51 are treated very differently
2. false negatives are dangerous because unsafe OOD samples may not be penalized

Our project asks whether a more conservative classifier objective and soft OOD usage can improve CCAC's safety signal.

Our Modification

We compare original CCAC with one selected variant.

Method	Classifier loss	OOD usage
Original CCAC	BCE	hard mask
Focal + soft	focal loss	soft OOD weight

Focal loss:

$$FL(p_t) = \alpha_t (1 - p_t)^\gamma BCE(p, y)$$

Soft OOD weighting:

$$w_{\text{OOD}}(s, a, \kappa) = \text{clamp}(h_\psi(s, a \mid \kappa), 0, 1)$$

Final variant: `classifier_loss=focal` , `ood_mode=soft` , `focal_alpha=0.75` , `focal_gamma=2.0` .

Experiment Design

We use a three-level validation pipeline instead of jumping directly to a full paper benchmark.

Test	Question	Cost
Test A: classifier-only	Does focal loss reduce unsafe OOD false negatives?	cheap
Test B: cost-critic separation	Does soft weighting strengthen OOD cost signal?	moderate
Test C: full policy	Does the mechanism transfer to policy behavior?	expensive

Scope:

- `OfflineBallRun-v0` : full policy comparison over seeds 0/1/2
- `OfflineCarRun-v0` : mechanism repeat over seeds 0/1 , policy seed 0
- all full policy runs use 50k update steps

Test A: OOD Classifier

False negative rate is the key safety metric: it measures unsafe/OOD samples incorrectly treated as in-distribution.

Task	Seed	Method	AUROC	AUPRC	FNR	FPR	ECE
BallRun	0	original	0.94270	0.93778	0.10380	0.17205	0.02227
BallRun	0	focal_soft	0.93943	0.93326	0.03565	0.28166	0.10002
CarRun	0	original	0.92558	0.88536	0.13206	0.17275	0.03152
CarRun	0	focal_soft	0.92649	0.88579	0.05261	0.28111	0.13474
CarRun	1	original	0.92868	0.89186	0.18108	0.12899	0.01671
CarRun	1	focal_soft	0.92992	0.89303	0.06185	0.25883	0.12755

Take-away: focal loss consistently lowers FNR, but increases FPR and worsens calibration.

Test A Interpretation

The classifier result supports a safety-oriented trade-off.

Positive signal

Focal loss misses fewer unsafe/OOD samples. This is useful because false negatives can prevent the cost critic from receiving conservative updates.

Cost of the change

The classifier becomes more conservative: FPR rises, ECE worsens, and the score should not be described as better calibrated.

Test B: Cost-Critic Separation

Test B compares generated OOD samples with matched in-distribution samples under the same query budget.

Task	Seed	Method	qc_matched_gap	qc_selected_matched_gap	OOD selected rate
BallRun	0	original	-26.829	88.324	0.4186
BallRun	0	focal_soft	8.711	6.131	0.7555
CarRun	0	original	2.641	4.929	0.3106
CarRun	0	focal_soft	3.632	6.644	0.4868
CarRun	1	original	3.267	6.069	0.2267
CarRun	1	focal_soft	5.031	8.027	0.4255

Test B Interpretation

The mechanism-level evidence is stronger than the policy-level evidence.

Observation	Meaning
More generated samples are selected or weighted as OOD	classifier becomes more conservative
Positive matched-budget gaps	OOD samples receive higher cost estimates than comparable IND samples
CarRun seed 0/1 both improve	mechanism result is not purely seed-specific
BallRun original has unstable scale	hard-mask behavior can produce noisy cost-critic separation

This supports the diagnostic claim: focal + soft OOD weighting improves the cost signal used by CCAC.

Test C: Full Policy Summary

Policy results are useful but mixed. `Safe targets` counts how many target costs in `[1, 2, 5, 10]` satisfy `real_cost <= target_cost`.

Task	Seed	Method	Avg reward	Avg real cost	Safe targets
BallRun	0	original	347.274	0.000	4/4
BallRun	0	focal_soft	422.771	0.000	4/4
BallRun	1	original	464.176	19.000	3/4
BallRun	1	focal_soft	420.406	3.250	4/4
BallRun	2	original	409.096	0.000	4/4
BallRun	2	focal_soft	443.294	9.650	3/4

BallRun: Useful but Not Uniform

Seed Main observation

0 `focal_soft` improves reward across all target costs while keeping zero cost.

1 `focal_soft` fixes the strict-budget failure: original has cost `74` at target `1`, while `focal_soft` has cost `0`.

2 `focal_soft` improves reward at target `2/5` and remains safe there, but fails at target `1` with cost `33.6`.

Conclusion for BallRun:

- the variant can improve reward or strict-budget safety in some settings
- it is not uniformly safer across seeds and budgets
- budget-conditioned policy stability remains incomplete

CarRun: Negative Policy Result

CarRun is the clearest policy-level failure case.

Method	Target 1 cost	Target 5 cost	Avg reward	Avg real cost	Safe targets
original	181.00	181.25	859.456	181.200	0/4
focal_soft	183.05	183.55	913.301	183.213	0/4

Interpretation:

- `focal_soft` increases reward on CarRun
- it does not reduce realized cost
- both methods remain far above all requested safety budgets after 50k updates

This shows that better OOD diagnostics do not automatically solve policy-level safe control.

Overall Findings

Level	Result	Interpretation
Classifier	FNR decreases on BallRun and CarRun	fewer unsafe/OOD samples are missed
Cost critic	matched-budget separation improves	stronger diagnostic OOD cost signal
BallRun policy	several useful improvements, one strict-budget failure	promising but unstable
CarRun policy	reward improves, cost remains unsafe	negative result

Final position:

`focal + soft` is a useful small-scale extension and diagnostic tool, not a robust new offline safe RL algorithm.

Limitations

1. Full policy training uses only 50k updates.
2. Policy-level evaluation covers BallRun seeds 0/1/2 and CarRun seed 0 .
3. Focal loss lowers FNR but worsens FPR and ECE.
4. Soft OOD weighting improves the cost signal but does not solve CarRun safety.
5. We do not compare against the full set of OSRL baselines.
6. The public DSRL server blocked the BallCircle fallback dataset during the final expansion.

These limits are important: the result is best read as a compact course-project extension with honest positive and negative evidence.

Conclusion

We extended CCAC with a small OOD regularization change:

- focal loss for the OOD classifier
- soft OOD score weighting in cost-critic updates

The extension improves mechanism-level diagnostics and some BallRun policy settings. However, full policy behavior is mixed, and CarRun remains unsafe.

The main contribution is a reproducible and interpretable analysis pipeline for OOD-based safety regularization in CCAC, with clear evidence about where the idea helps and where it still fails.

Q&A

Thank you!

References

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